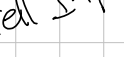


$aba^{-1}b^{-1} = e$
 $ab = ba \rightarrow \text{group}$
 $S^1 \vee S^2$
 $T = \mathbb{R} \times \mathbb{R}$
 $T \hookrightarrow \mathbb{R}^2$
 $\mathbb{R} \hookrightarrow \mathbb{R}^2$

[illegible]

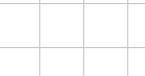

 $S: 0\text{-cell } \downarrow$
 $1\text{-cell } \downarrow$


 $S: 0\text{-cell } \downarrow$
 $2\text{-cell } \downarrow$


 $S' \vee S'$

$R/P:$

$0\text{-cell: } \downarrow$
 $1\text{-cell: } \downarrow$
 $2\text{-cell } \downarrow$



$ab^{-1}b^{-1} = 0$
 $ab = b^{-1}$

Handwritten sketches and notes:

- A circle containing a horizontal wavy line.
- A flower-like shape with five petals.
- A small figure-eight or infinity symbol.
- A vertical line segment with a dot at its top end.
- The word "Kugel" (German for sphere) written twice.
- A coordinate point $(0,0,2)$.
- A square with dots at each corner.
- Mathematical expressions: $\frac{1}{2}, \frac{1}{2}, \frac{1}{2}$ and $\frac{1}{2}!$.